P001-MVB

Responsible AI Clinical Decision Support Prototype

Explainable and Responsible AI for Differentiating Bacterial and Viral Meningitis: A Clinical Decision Support System

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Programme
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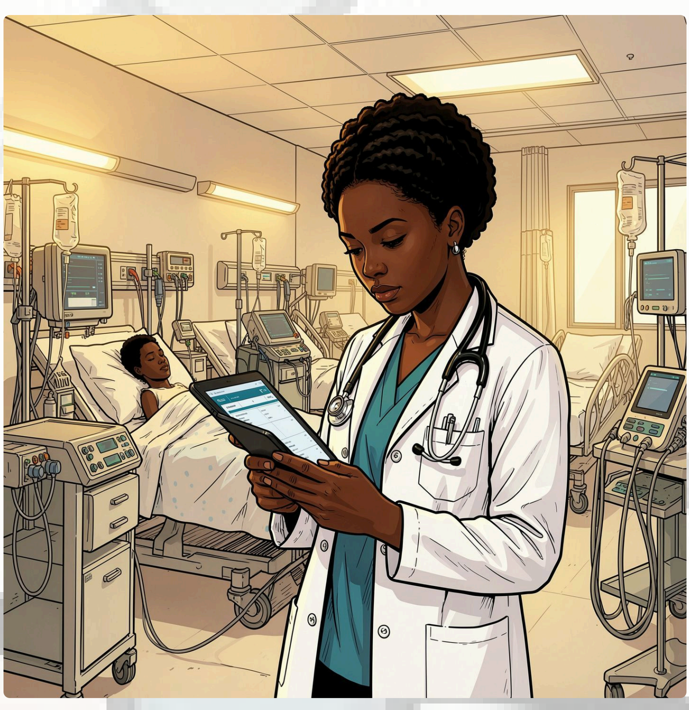
Location
Abuja, Nigeria

Accurate Prediction | Explainable AI | Responsible Governance | Human Oversight

The Problem



- ⚠️ **Delayed Diagnosis** — Rapidly distinguishing **bacterial** from **viral meningitis** is clinically challenging, yet treatment decisions must often be made **urgently**.
- ⚠️ **Black-Box AI** — Many healthcare AI systems operate as **black boxes**, offering little explanation for their predictions.
- ⚠️ **Limited Transparency** — This lack of transparency reduces clinician trust and confidence, increases automation bias, and limits safe adoption in healthcare.

Our Solution — P001-MVB

P001-MVB integrates four complementary components delivered through an interactive **Streamlit** clinical decision-support application, prioritising transparency, explainability, governance, and mandatory human oversight.

Explainable Machine Learning
Logistic Regression with interpretable outputs

SHAP Explanations
Feature-level transparency for every prediction

Responsible AI Governance
Model Card, AI Risk Assessment, structured governance

PDVC
Pre-Decision Verification Checkpoint — mandatory clinician review

P001-MVB Clinical Decision Workflow

1. **Patient Labs** — Collect blood and CSF results
2. **ML Prediction** — Model estimates bacterial vs viral
3. **SHAP Explanation** — Feature contributions clarified
4. **PDVC Checkpoint** — Privacy, data validity, compliance
5. **Final Clinical Decision** — Expert assesses model output

Every prediction passes through explainability and a mandatory verification checkpoint before influencing any clinical decision — supporting, not replacing, clinical judgement.

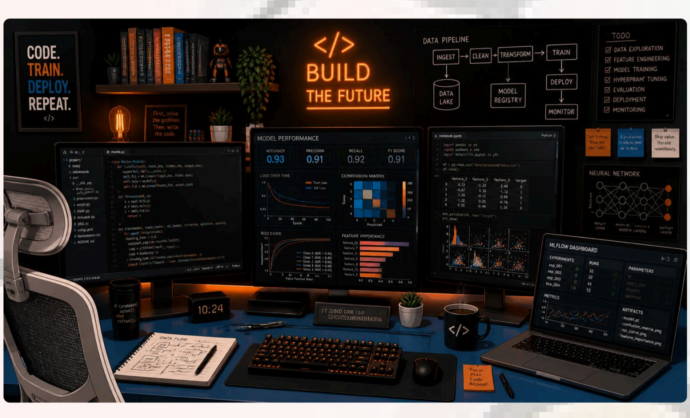
Tools & Technologies

Programming & ML
Python · Pandas · NumPy · Scikit-learn · XGBoost · SHAP

Deployment
Streamlit Community Cloud · GitHub

Development
Google Colab · Jupyter Notebook

Responsible AI
Model Card · AI Risk Assessment · PDVC Framework



End-to-End AI Development Lifecycle

Proposal → Data Acquisition → Data Preparation → Exploratory Analysis → Feature Engineering

Six machine learning models were evaluated. The final model was selected based on **Bacterial Recall** — prioritising patient safety over overall accuracy.

Near-Proxy Removal
Removed **Pathogen_Present** feature to prevent data leakage

Feature Set Comparison
Full vs Restricted feature sets rigorously evaluated

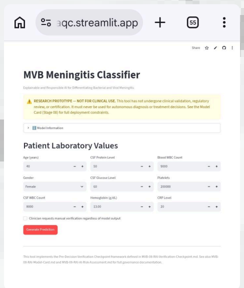
Safety-First Model Selection
Bacterial Recall prioritised — minimising missed diagnoses

Responsible AI Governance
Embedded governance integrated directly into the deployed application

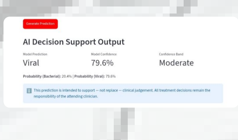
Results & Live Application Demonstration

Final Selected Model: Logistic Regression (Restricted Feature Set)

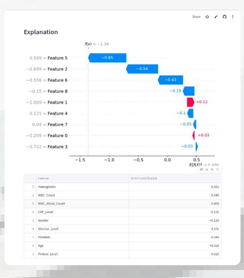
Streamlit Home



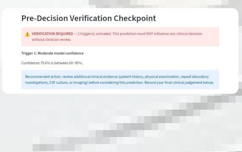
Prediction Result



SHAP Waterfall



PDVC Verification



Pre-Decision Verification Checkpoint (PDVC) automatically requiring clinician review when governance triggers are activated, with an audit log supporting transparency and accountability.

93.4%
Accuracy
Overall model accuracy on held-out test data

95.8%
Bacterial Recall
Critical safety metric — minimising missed bacterial cases

91.9%
Precision
Proportion of bacterial predictions that are correct

5
PDVC Triggers
Automated governance triggers plus clinician-initiated escalation

Responsible AI Outcomes

- SHAP provides transparent **feature-level explanations** for every prediction
- All predictions include **confidence scores**
- All five false-negative evaluation cases would have **triggered PDVC verification** before influencing clinical decisions

All five false-negative cases would have been flagged by the PDVC — demonstrating the safety net in practice.

Live Application: Screenshots of the Streamlit Home Screen, Prediction Results, SHAP Explanation, and PDVC Verification screens are available in the live demonstration.

Pre-Decision Verification Checkpoint (PDVC)

Many healthcare AI projects stop after producing an accurate prediction. **P001-MVB** goes further by integrating Responsible AI governance directly into the clinical decision-support workflow.

Detects
Higher-risk predictions automatically

Requires
Mandatory clinician verification before action

Promotes
Transparency through embedded explainability

Reduces
Automation bias and over-reliance on AI

Reinforces
Clinician authority and accountability

> **Governance is not documentation—it is an operational safeguard embedded in the deployed AI system.**

Conclusion & Future Work

→ **Key Takeaway 1**
Explainable AI improves trust by making each prediction transparent to clinicians.

→ **Key Takeaway 2**
Safety-first model selection prioritises bacterial recall to reduce missed critical cases.

→ **Key Takeaway 3**
Responsible AI governance is embedded directly into the workflow through PDVC.

→ **Key Takeaway 4**
Human oversight remains central, supporting clinical judgement rather than replacing it.

01
Expand Dataset Coverage
Collect additional clinical cases to improve robustness and generalisability.

02
Validate Across Sites
Test performance with data from multiple hospitals and care settings.

03
Refine Governance Rules
Enhance PDVC triggers, thresholds, and escalation logic for safer deployment.

04
Improve Model Performance
Compare additional interpretable models and calibration strategies.

05
Prepare Clinical Integration
Develop workflow-ready deployment features for real-world use.

Live Resources

GitHub Repository

https://github.com/Abamid/P001-MVB

Live Streamlit Application

https://p001-mvb-iq6ziqmh6yr9tbdamggaqc.streamlit.app/

Scan the QR codes to explore the complete project.

Thank You

Questions?

Abubakar Amidu · 3MTT DeepTech Cohort 2 – DS/ML Mentorship · Abuja, Nigeria

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